Extending BESSTIE: Prompt-Based vs Fine-Tuned Models for Dialectal Sentiment Classification

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***Abstract*—Large language models have been shown to exhibit performance disparities across varieties of English. The BESSTIE benchmark formalizes this issue for sentiment and sarcasm classification across Australian, Indian, and British English. In this report, we extend the BESSTIE evaluation in two directions. First, we conduct a systematic comparison between encoder-based fine-tuned models and prompt-based inference under zero-shot and supervised settings for dialectal sentiment classification. Second, we study sarcasm detection on a filtered BESSTIE subset (Sarcasm, Reddit) and evaluate whether data augmentation for class balancing and supervised contrastive learning improve minority-class recognition. Our findings confirm uneven gains across varieties in Extension 1, and show that in Extension 2 augmentation substantially improves macro-level balance by increasing recall on sarcastic examples, while contrastive training provides additional but hyperparameter-sensitive benefits.**

1 Introduction

Despite strong performance on Standard English, large language models (LLMs) often struggle with non-standard language varieties. Prior work has shown that dialectal variation can introduce systematic bias in NLP systems, particularly for subjective tasks such as sentiment and sarcasm classification.

The BESSTIE benchmark addresses this gap by providing annotated datasets for three varieties of English: Australian (en-AU), Indian (en-IN), and British (en-UK). While the original work focuses on fine-tuning both encoder and decoder models, several open questions remain regarding the comparative behavior of prompt-based versus fine-tuned approaches.

This report presents an extension of BESSTIE with two goals: (1) to empirically analyze performance differences between fine-tuned encoder models and prompt- based approaches under a unified experimental setup for dialectal sentiment classification, and (2) to investigate a concrete training-based extension for sarcasm detection on BESSTIE by combining class-balancing data augmentation with supervised contrastive learning, aiming to improve the detection of minority sarcastic instances.

2 Background: The BESSTIE Benchmark

BESSTIE is a benchmark for sentiment and sarcasm classification across English varieties, constructed from Google Places reviews and Reddit comments. The dataset is manually annotated by native speakers and validated using both manual and automatic variety identification techniques.

The original study evaluates nine models spanning encoder and decoder architectures. Results indicate consistently higher performance on inner-circle varieties (en-AU, en-UK) compared to the outer-circle variety (en-IN), with sarcasm classification posing a greater challenge than sentiment analysis.

2.1 Summary of the Base Paper

BESSTIE is a benchmark for sentiment and sarcasm classification across three English varieties: Australian (en-AU), Indian (en-IN), and British (en-UK), addressing the lack of variety-aware evaluation in standard NLP benchmarks. It contains manually annotated data from two domains: Google Places reviews and Reddit comments, collected via location/topic filtering and validated through native-speaker annotation and automatic variety identification. The base paper evaluates nine encoder and decoder LLMs: encoders are fine-tuned with supervised classification heads, while decoders are tested via instruction-style prompting. Results are reported using macro-F1 and show systematic disparities, with higher performance on en-AU/en-UK than on en-IN (especially for sarcasm), and limited cross-variety generalization even after fine-tuning.

2.2 Key Results from BESSTIE

The BESSTIE benchmark reveals consistent performance disparities across English varieties for both sentiment and sarcasm classification. Averaged across all models, sentiment classification achieves higher performance than sarcasm classification, indicating that sarcasm remains a substantially harder task.

Across all evaluated models, inner-circle varieties (en-AU and en-UK) achieve higher macro-averaged F1 scores than the outer-circle variety (en-IN). This gap is most pronounced for sarcasm classification on Reddit data, where performance drops substantially for en-IN. Even for sentiment classification, models show uneven gains from fine-tuning, benefiting en-AU and en-UK more than en-IN.

Finally, cross-variety evaluation demonstrates limited generalization, particularly for sarcasm detection. Fine-tuning

Model	Sent. (G)	Sent. (R)	Sarc. (R)
ALBERT	0.88	0.75	0.65
BERT	0.88	0.80	0.60
RoBERTa	0.89	0.79	0.55
mBERT	0.88	0.65	0.68
mDistilBERT	0.86	0.78	0.51
XLM-R	0.77	0.73	0.62
Gemma	0.79	0.60	0.45
Mistral	**0.91**	**0.84**	**0.68**
Qwen	0.45	0.82	0.52

TABLE I: Model-wise macro-F1 scores averaged across English varieties, as reported in the BESSTIE benchmark.

improves in-variety performance but often degrades cross-variety performance, highlighting the importance of variety-specific data and evaluation.

3 Extension 1: Prompt-Based vs Fine-Tuned Models

3.1 Motivation

While BESSTIE evaluates both encoder and decoder models, it does not directly contrast prompt-based inference with supervised fine-tuning under a unified setup. This extension isolates that comparison to understand how training paradigms interact with language variety.

3.2 Setup and Evaluation

We reuse the BESSTIE splits and metrics for sentiment classification across en-AU, en-IN, and en-UK. Encoder-based models (BERT, RoBERTa, ALBERT, DeBERTa-base) are fine-tuned with a standard classification head. Prompt-based approaches are evaluated in zero-shot settings, including NLI-style inference and instruction/pseudo zero-shot prompting. All approaches are evaluated on the BESSTIE validation split using macro-F1. For fine-tuned encoders, predictions are obtained via argmax over the two logits; for NLI-based models, we map the highest-scoring label among {*negative*, *positive*} to the corresponding class. We report both per-variety scores and their mean to expose dialect-wise differences that may be hidden by aggregate metrics.

3.3 Results and Discussion

Table II reveals a consistent ranking when averaged across varieties: **NLI-based zero-shot** $>$ **fine-tuned encoders** $>$ **instruction / pseudo zero-shot prompting**. Fig.1 visualizes the same trend across dialects, highlighting the stronger robustness of NLI-based zero-shot models and the larger degradation on en-IN for prompt-only approaches. Fine-tuned encoders reach competitive in-variety macro-F1, especially on en-AU and en-UK, yet the gains are less consistent for en-IN. In contrast, NLI-based zero-shot models obtain a higher mean macro-F1 and display more stable behavior across varieties without any task-specific fine-tuning, suggesting that broad semantic inference training can yield more transferable decision mechanisms under dialectal variation. To quantify whether this gap is robust

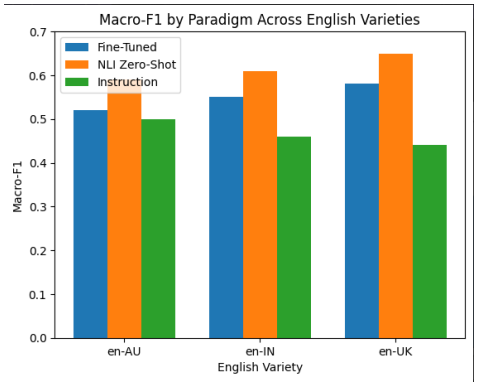


Fig. 1: Macro-F1 across English varieties for the three main paradigms.

rather than a consequence of random variation, we performed a **Welch's independent** t**-test**, treating per-variety macro-F1 scores as independent observations ($n = 12$ per group). We adopt Welch's test because it does not require the assumption of equal variances between groups, which is appropriate when different model families may exhibit heterogeneous variability across dialects and domains. The test indicates that NLI-based zero-shot models significantly outperform fine-tuned encoders ($t = -2.18$, $p = 0.046$). Alongside statistical significance, we report **Cohen's** d as an effect-size measure to express the magnitude of the difference in standardized units; the resulting value ($d = 0.89$) corresponds to a large effect, supporting the interpretation that the advantage is not only detectable but also substantial.

Finally, pseudo zero-shot and instruction prompting underperform relative to both fine-tuned and NLI-based models, particularly on en-IN, highlighting the limits of prompt-only adaptation when dealing with pragmatic variation and lexical shifts.

Model	Paradigm	en-AU	en-IN	en-UK
ALBERT-base-v2	Fine-tuned	0.57	0.61	0.65
BERT-base-uncased	Fine-tuned	0.55	0.56	0.63
DeBERTa-v3-base	Fine-tuned	0.38	0.42	0.41
RoBERTa-base	Fine-tuned	0.58	0.62	0.64
BART-large-MNLI	NLI zero-shot	0.61	0.63	0.66
DeBERTa-v3-large (Laurer)	NLI zero-shot	0.58	0.63	0.65
RoBERTa-large-MNLI	NLI zero-shot	0.61	0.61	0.66
XLM-R-large-XNLI	NLI zero-shot	0.57	0.57	0.63
DeBERTa-v3-large (vanilla)	Pseudo zero-shot	0.52	0.45	0.47
FLAN-T5-large	Instruction	0.47	0.46	0.41

TABLE II: Macro-F1 performance across English varieties.

3.4 Takeaways

Overall, stronger supervision does not necessarily translate into stronger cross-variety robustness. NLI-based prompting provides a competitive and robust alternative in dialect-sensitive settings, whereas instruction prompting alone remains insufficient to compensate for variety-specific differences.

4 Extension 2: Sarcasm-only Training with Data Augmentation and Supervised Contrastive Learning

4.1 Motivation

BESSTIE highlights that subjective NLP tasks are sensitive to language variation and domain effects, and that sarcasm detection is particularly challenging due to its pragmatic nature and reliance on implicit cues. In this extension, we focus on the *Sarcasm* task and explore whether changing the training strategy (rather than changing the model family) can yield more robust decision boundaries. Specifically, we study two complementary ideas: (i) **data augmentation** to mitigate class imbalance and increase surface-form diversity, and (ii) **supervised contrastive learning** to shape the embedding space so that sarcastic vs. non-sarcastic examples become more separable.

4.2 Dataset Subset and Experimental Protocol

We load BESSTIE from HuggingFace and filter the dataset to keep **only** examples with **Sarcasm** task on **Reddit** source. This makes the experimental setting coherent (no mixing with sentiment examples) and aligns with the observation that sarcasm is commonly studied in conversational and social media contexts. After filtering, the training split contains 4,480 examples and the validation split 612 examples, with a noticeable class imbalance (non-sarcastic as majority class). Fig.2 visualizes the resulting class imbalance.

All models are evaluated on the same validation split using **macro-F1** as the primary metric, complemented by accuracy, macro-precision, and macro-recall. Macro-F1 is preferred because it weights classes equally and therefore better reflects performance on the minority sarcastic class.

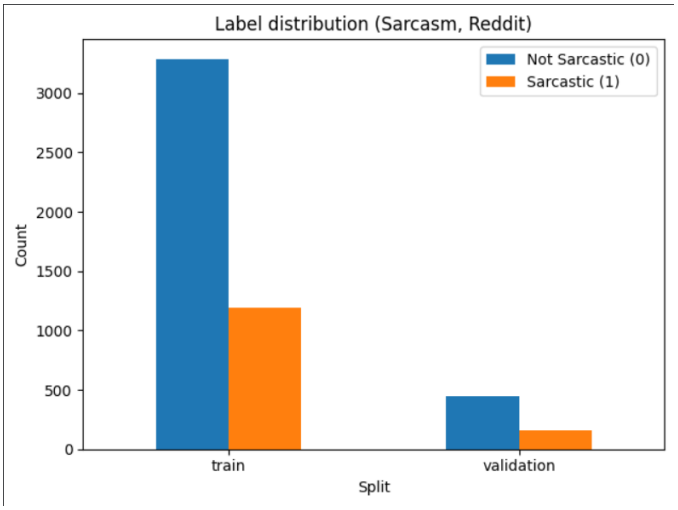


Fig. 2: Class distribution on the filtered Sarcasm-Reddit subset

4.3 Methods

All approaches use the same backbone encoder, **XLM-RoBERTa-base**, and differ only in the training recipe.

4.3.1 Baseline: Cross-Entropy finetuning: As a reference system, we fine-tune `xlm-roberta-base` with a standard sequence classification head (two output logits) and optimize cross-entropy loss on the filtered sarcasm dataset.

4.3.2 (Augmented Baseline: Data Augmentation + Cross-Entropy: To reduce class imbalance and increase robustness to surface variation, we construct an augmented training set with two steps: (i) *class balancing*: we oversample the minority class by generating augmented variants of sarcastic examples until both classes have comparable counts; and (ii) *diversity augmentation*: we additionally augment each training example once to increase lexical and orthographic variability. The augmentation operators are intentionally lightweight (to preserve semantics), including spelling swaps (e.g., British/American variants), contraction expansion/compression, and small punctuation/noise perturbations. Negation-sensitive words (e.g., *not*, *never*) are protected to avoid label-flipping artifacts. We then fine-tune the same backbone on this larger augmented dataset using cross-entropy.

4.3.3 Supervised Contrastive Learning + Cross-Entropy (SupCon+CE): The core extension introduces a contrastive objective on top of the encoder representations. We build a dual-head architecture: (1) a projection head mapping the CLS embedding to a normalized vector for contrastive learning, and (2) a standard classification head for sarcasm prediction. Training minimizes a weighted combination of supervised contrastive loss and cross-entropy:

$$\mathcal{L} = \alpha\, \mathcal{L}_{\text{SupCon}} + (1-\alpha)\, \mathcal{L}_{\text{CE}}, \tag{1}$$

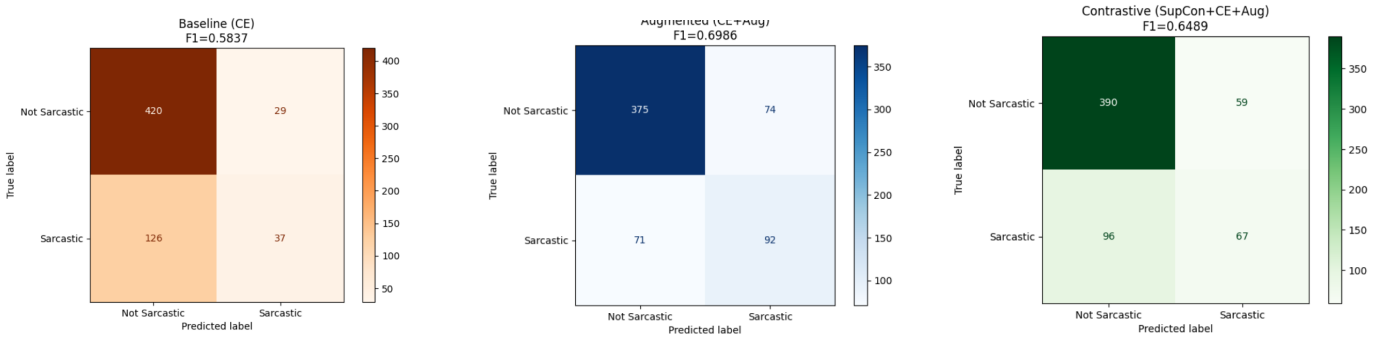
where α controls the trade-off. Intuitively, $\mathcal{L}_{\text{SupCon}}$ encourages examples with the same label to cluster in embedding space, while separating different-label examples, potentially improving class separability and minority-class recall. In our implementation we set $\alpha = 0.3$ and a contrastive temperature $\tau = 0.07$. This model is trained on the same augmented dataset as in (2).

4.4 Implementation Details

Tokenization uses a maximum length of 256 tokens with dynamic padding. Training is performed with the HuggingFace `Trainer` API and checkpointing to support resumable runs. We use AdamW with learning rate $2{\cdot}10^{-5}$, weight decay 0.01, batch size 16 for training and 32 for evaluation. The baseline and augmented baseline are trained for 3 epochs, while the contrastive model is trained for 4 epochs with a warmup ratio of 0.1. The best checkpoint is selected by validation macro-F1.

Model	$\mathbf{F1_{macro}}$	**Acc.**	$\mathbf{Prec_{macro}}$	$\mathbf{Rec_{macro}}$
Baseline (CE)	0.584	0.747	0.665	0.581
Augmented (CE+Aug)	**0.699**	**0.763**	0.698	**0.700**
Contrastive (SupCon+CE+Aug)	0.649	0.747	0.667	0.640

TABLE III: Extension 2 results on the sarcasm-only Reddit subset.



(a) Baseline (CE) (b) Augmented (CE+Aug) (c) Contrastive (SupCon+CE+Aug)

Fig. 3: Confusion matrices on the sarcasm-only Reddit validation set for the three training strategies.

4.5 Results and Analysis

Table III summarizes the main results. The augmented baseline (CE+Aug) achieves the highest macro-F1, while the contrastive variant improves over the plain CE baseline but remains below CE+Aug in this configuration.

4.5.1 Macro-F1 vs. Accuracy: why the baseline can look deceptively good: Although the baseline accuracy is relatively high, it is largely driven by the majority class (non-sarcastic). In practice, the baseline tends to miss a substantial portion of sarcastic instances, which lowers minority-class recall. Macro-F1 is therefore more informative here because it penalizes models that perform well on the majority class while under-detecting sarcasm.

4.5.2 Error profile from confusion matrices: what changes with augmentation.: Figure 3 shows that augmentation shifts the error profile in a predictable way: the model becomes less conservative in predicting sarcasm. This reduces false negatives on sarcastic examples (higher sarcastic recall), but increases false positives (more non-sarcastic comments flagged as sarcastic). This FP–FN trade-off explains why macro-F1 improves even when accuracy changes only moderately.

4.5.3 Role of supervised contrastive learning: partial recovery and sensitivity.: The contrastive model yields an intermediate behavior: it preserves part of the recall gain obtained with augmentation while reducing some of the additional false positives, but it does not surpass CE+Aug overall. This suggests that the contrastive objective can help shape representations, yet its benefit is sensitive to the setup (e.g., α, temperature τ, batch composition, or negative sampling), and may require systematic tuning to add consistent value beyond a strong augmented cross-entropy baseline.

4.6 Limitations and Future Work

This extension currently evaluates performance aggregated over the filtered validation split. Since BESSTIE is fundamentally cross-variety, an important next step is to report macro-F1 separately for `en-AU`, `en-IN`, and `en-UK` within the sarcasm-only subset, and to test whether augmentation and contrastive learning reduce or amplify variety gaps. Additionally, contrastive learning performance may be sensitive to hyperparameters and batch sampling; future work should explore systematic tuning of α and τ, larger effective batch sizes (or memory banks), and augmentation strategies that are explicitly variety-aware rather than purely surface-level.

5 Overall Conclusion and Future Work

We presented two complementary extensions of BESSTIE. Extension 1 compared fine-tuned encoders with prompt-based and NLI-style zero-shot inference for dialectal sentiment classification, showing that stronger aggregate results can still mask non-uniform gains across varieties. Extension 2 focused on sarcasm on a filtered subset (Sarcasm, Reddit) and showed that class-balancing augmentation improves macro-F1 mainly by increasing recall on sarcastic instances, while the contrastive setup provides smaller, configuration-sensitive gains. Future work includes cross-variety transfer evaluations for Extension 1 and per-variety macro-F1 on the sarcasm-only subset for Extension 2, along with variety-aware augmentation and systematic tuning of the contrastive setup.